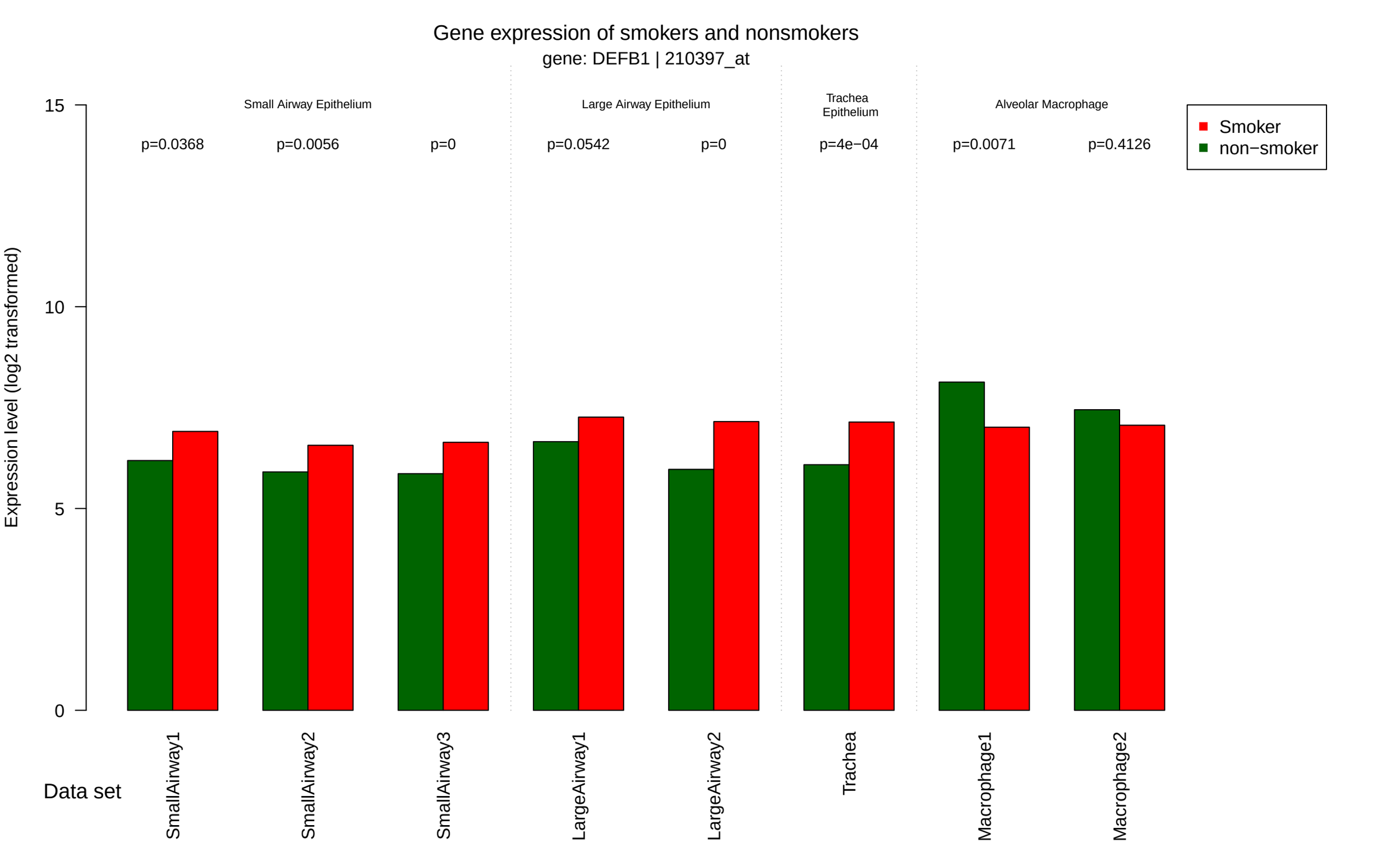
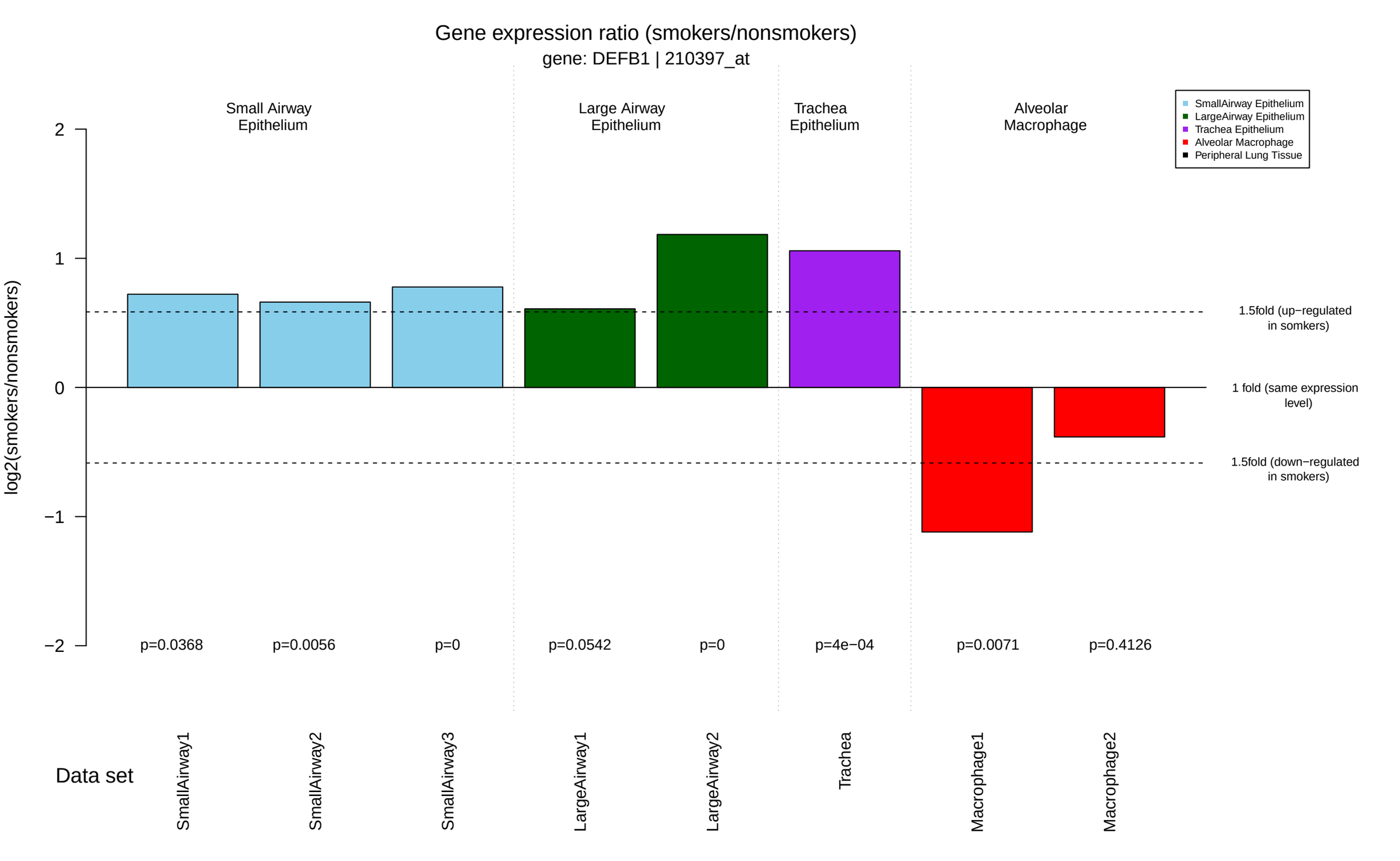
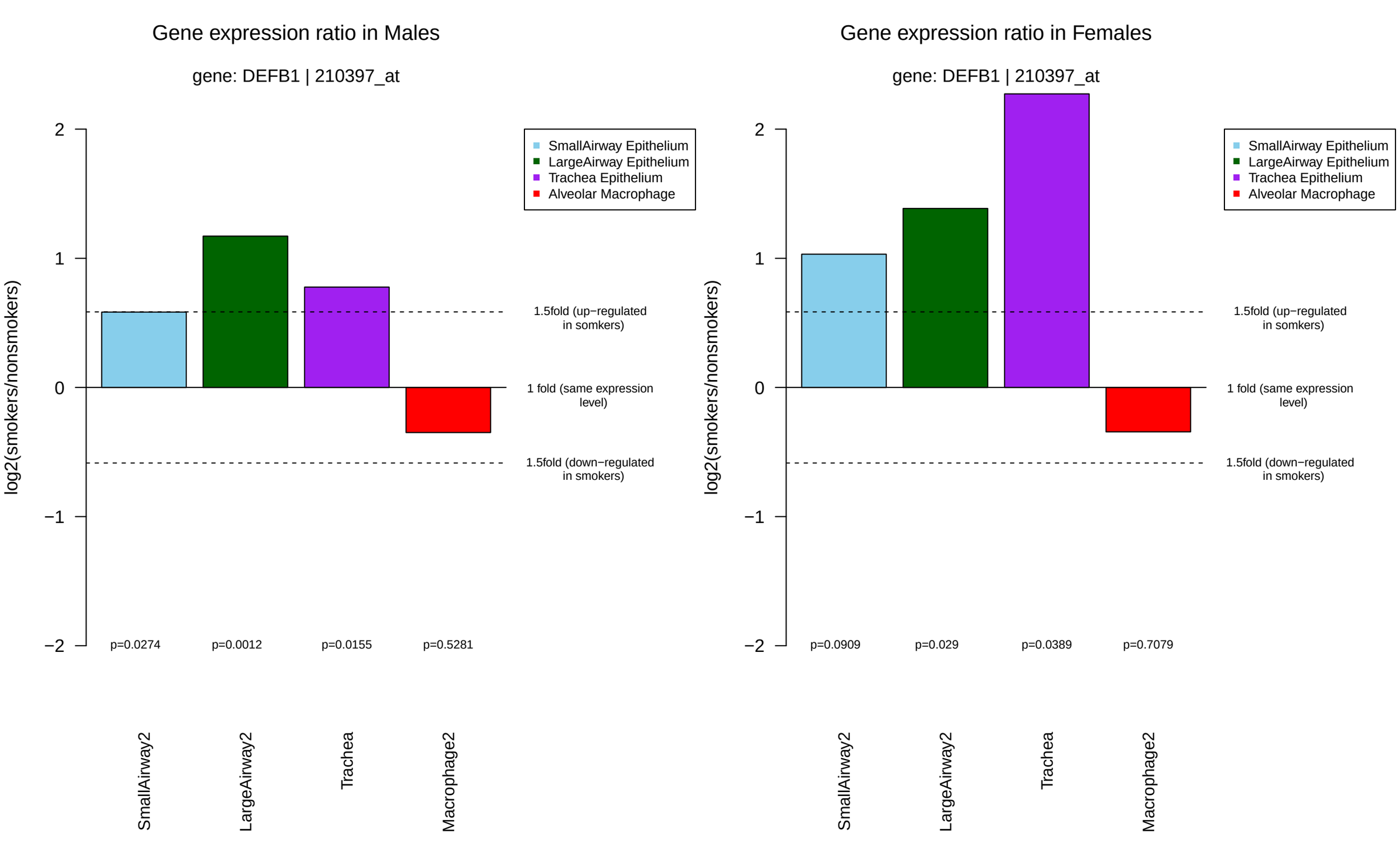
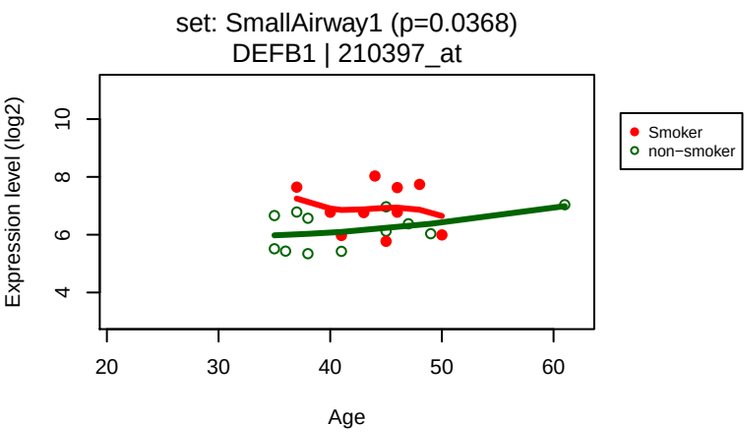
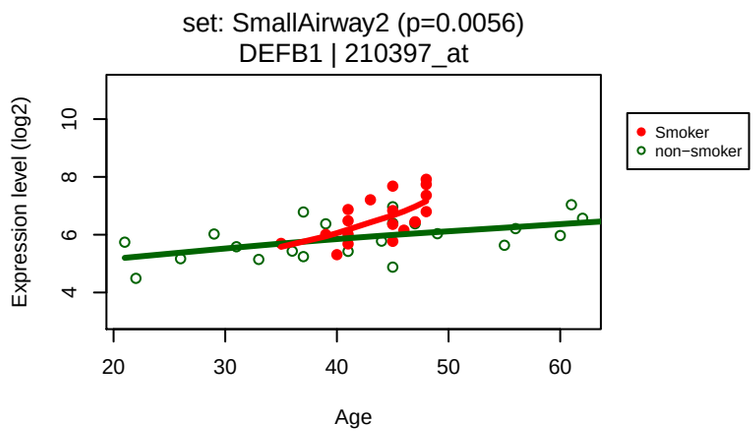
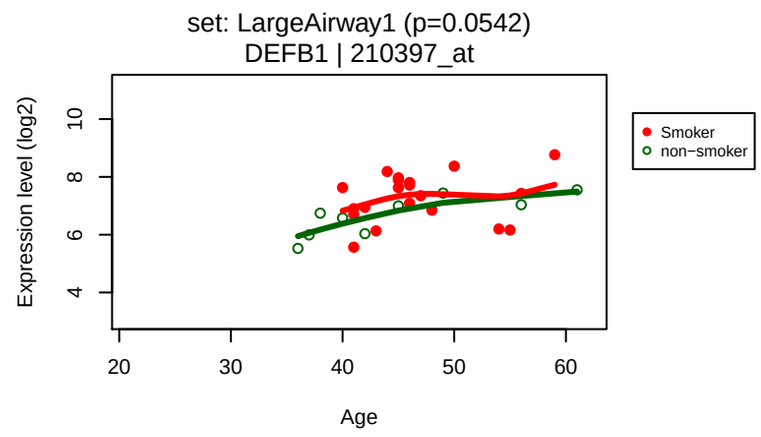

Gene expression of smokers and nonsmokers
gene: DEFB1 | 210397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0368
p=0.0056
p=0
p=0.0542
p=0
p=4e−04
p=0.0071
p=0.4126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

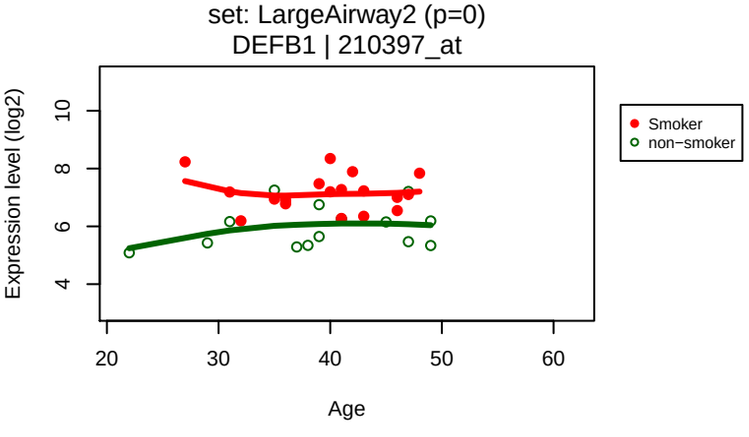

Gene expression ratio (smokers/nonsmokers)
gene: DEFB1 | 210397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0368
p=0.0056
p=0
p=0.0542
p=0
p=4e−04
p=0.0071
p=0.4126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

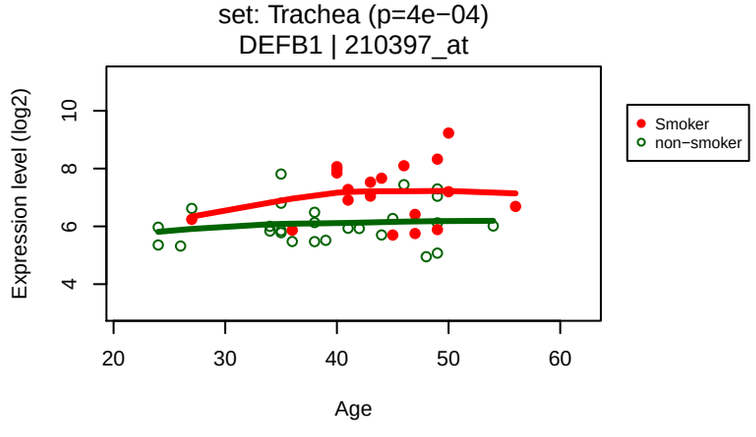

Gene expression ratio in Males
gene: DEFB1 | 210397_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0274
p=0.0012
p=0.0155
p=0.5281
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DEFB1 | 210397_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0909
p=0.029
p=0.0389
p=0.7079
SmallAirway2
LargeAirway2
Trachea
Macrophage2

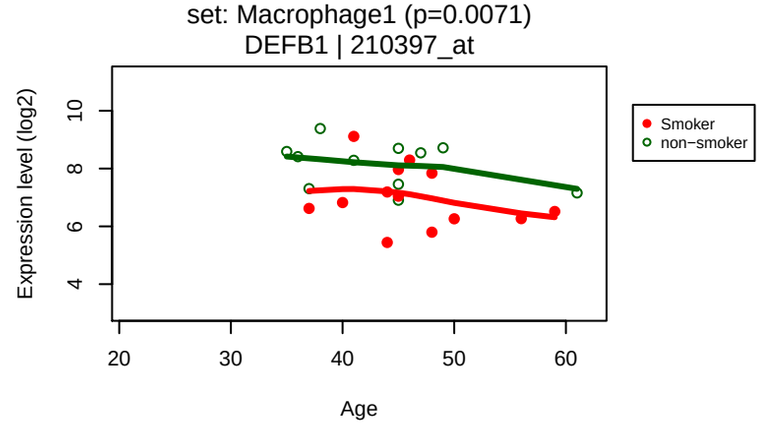

set: SmallAirway1 (p=0.0368)
DEFB1 | 210397_at
Expression level (log2)
Age
Smoker
non−smoker

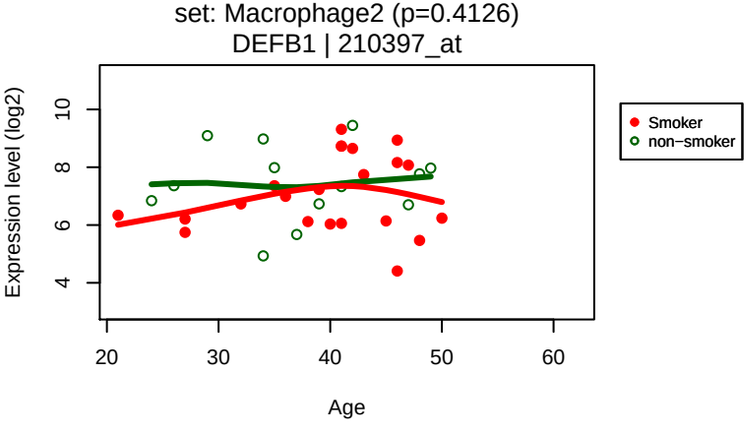

set: SmallAirway2 (p=0.0056)
DEFB1 | 210397_at
Expression level (log2)
Age
Smoker
non−smoker

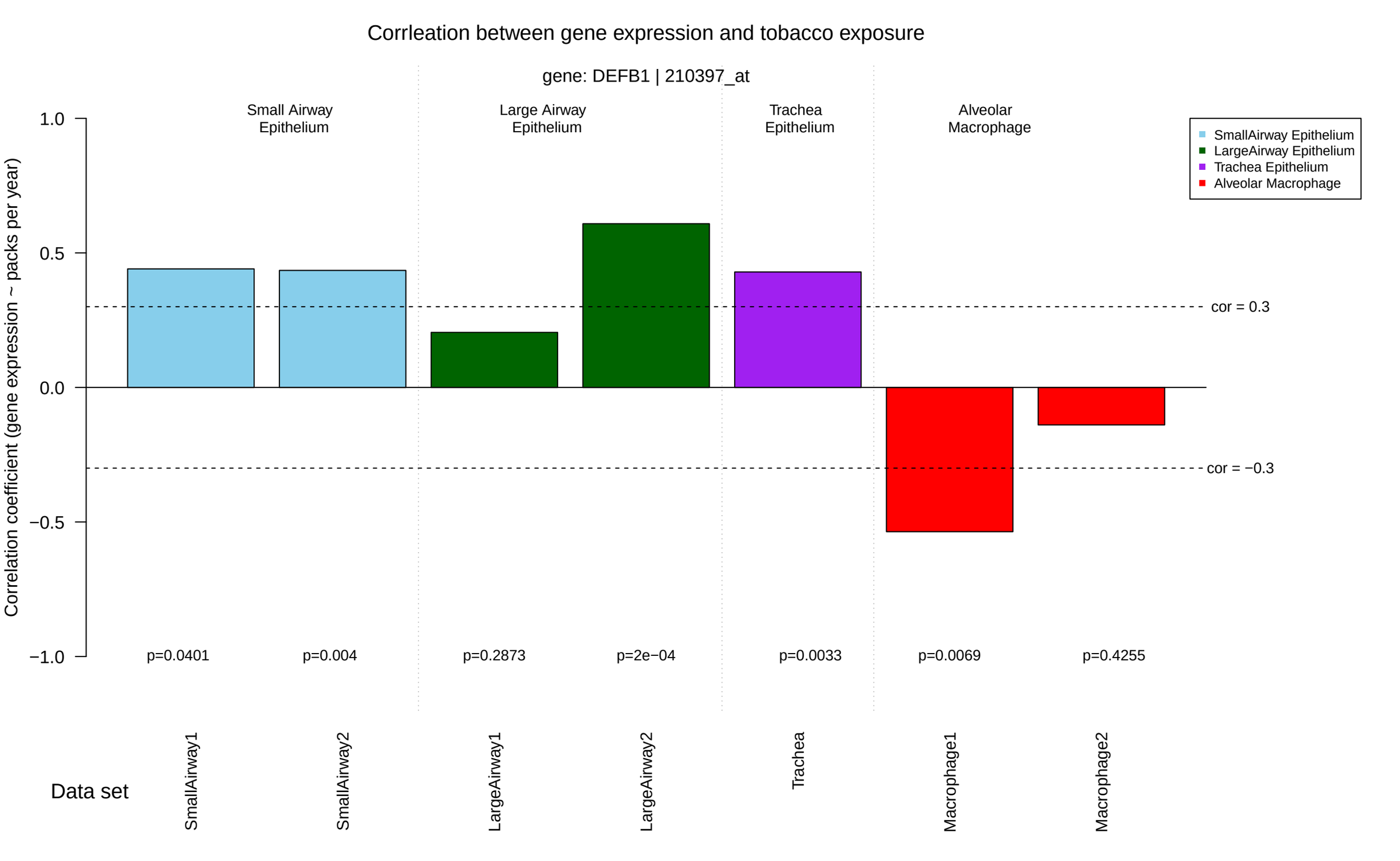

set: LargeAirway1 (p=0.0542)
DEFB1 | 210397_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0)
DEFB1 | 210397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
DEFB1 | 210397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0071)
DEFB1 | 210397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4126)
DEFB1 | 210397_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DEFB1 | 210397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0401
p=0.004
p=0.2873
p=2e−04
p=0.0033
p=0.0069
p=0.4255
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2